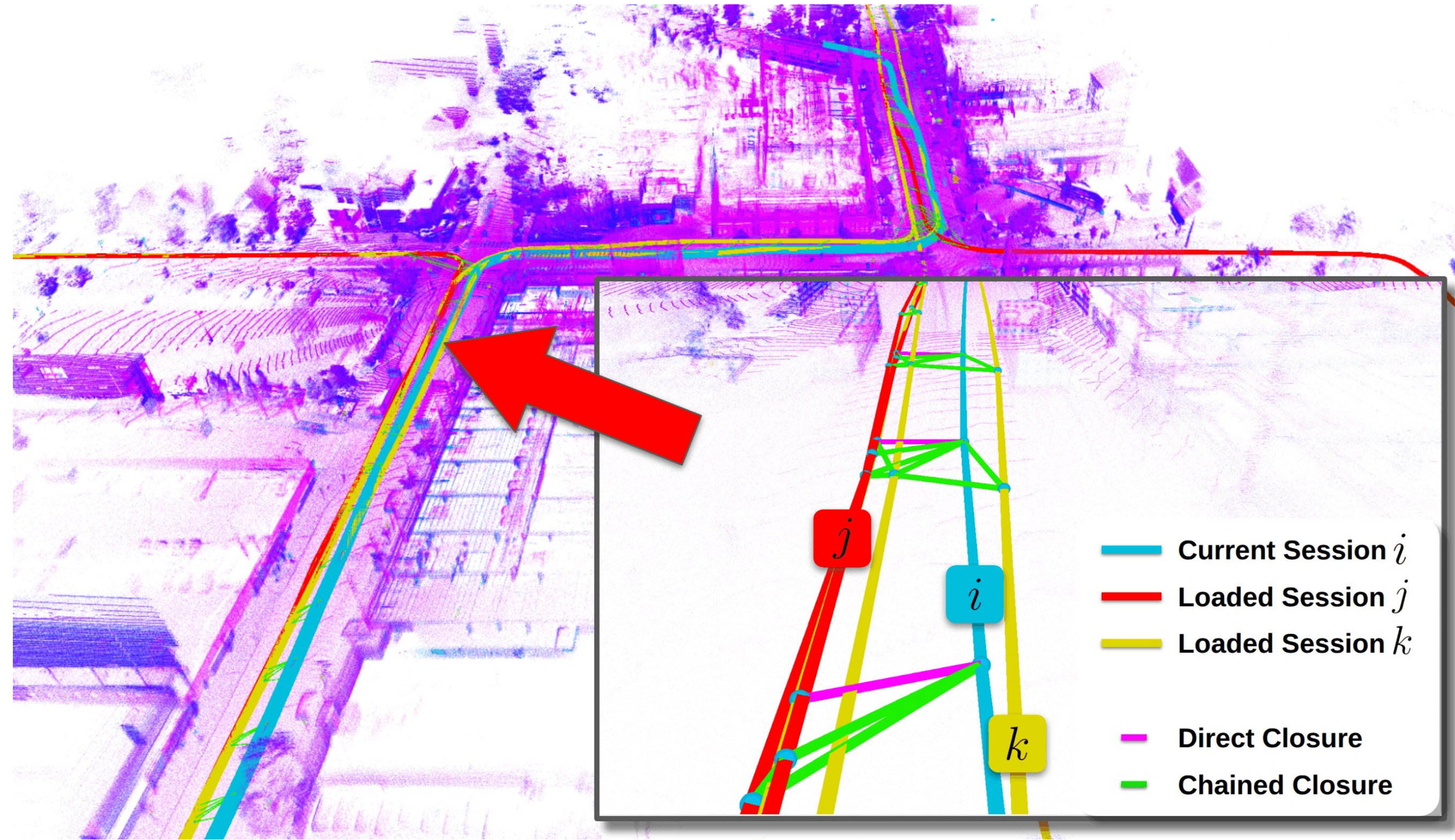




Motivation



Direct and chained closures link live session i to loaded sessions j, k .

Large environments span **many sessions**, and long time horizons change their appearance. **Robust multi-session loop closure** is essential.

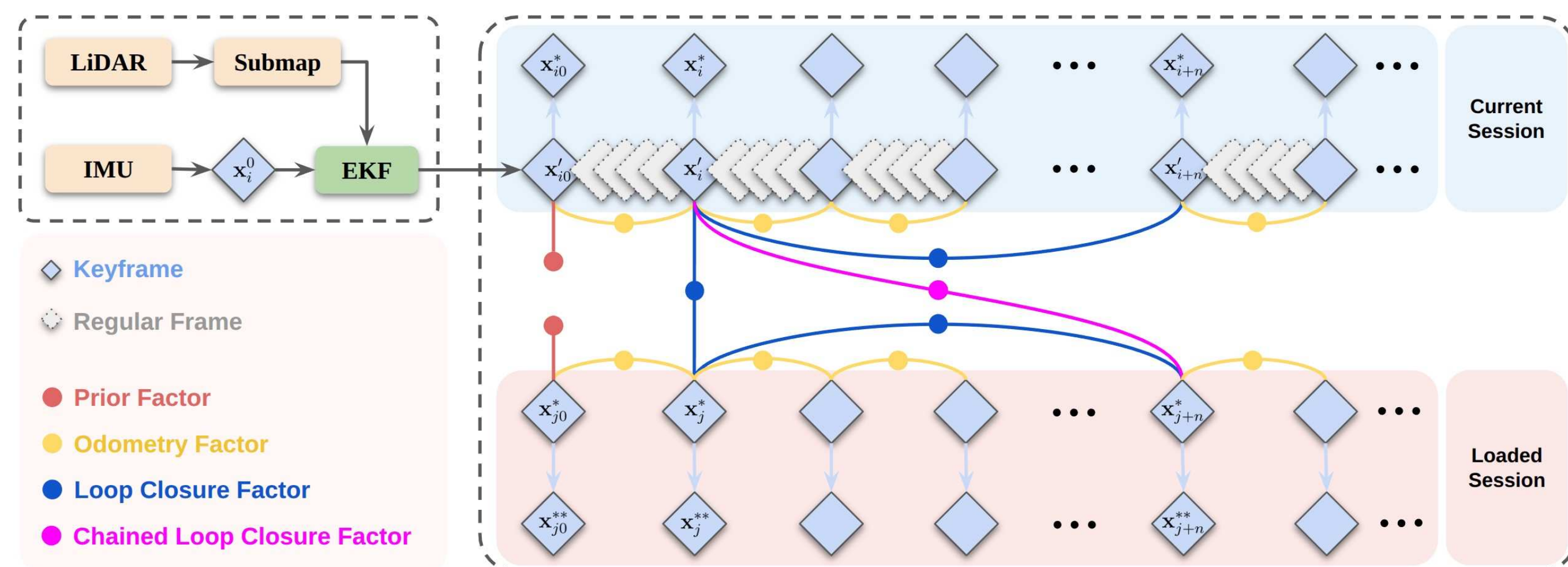
Limitations of existing methods

- **Post-processing**: map merged offline after sessions finished.
- **Limited scale**: a global solve over many closures struggles to converge.
- **Short temporal horizon**: scene change degrades closure detection.

Contributions

- **Online multi-session map reuse**: prior maps join the factor graph, aligned on-the-fly and re-optimized in iSAM2.
- **Efficient long-horizon alignment**: via chained loop closure.
- **Robustness**: one hyperparameter set, minimum tuning across platforms.

System Overview



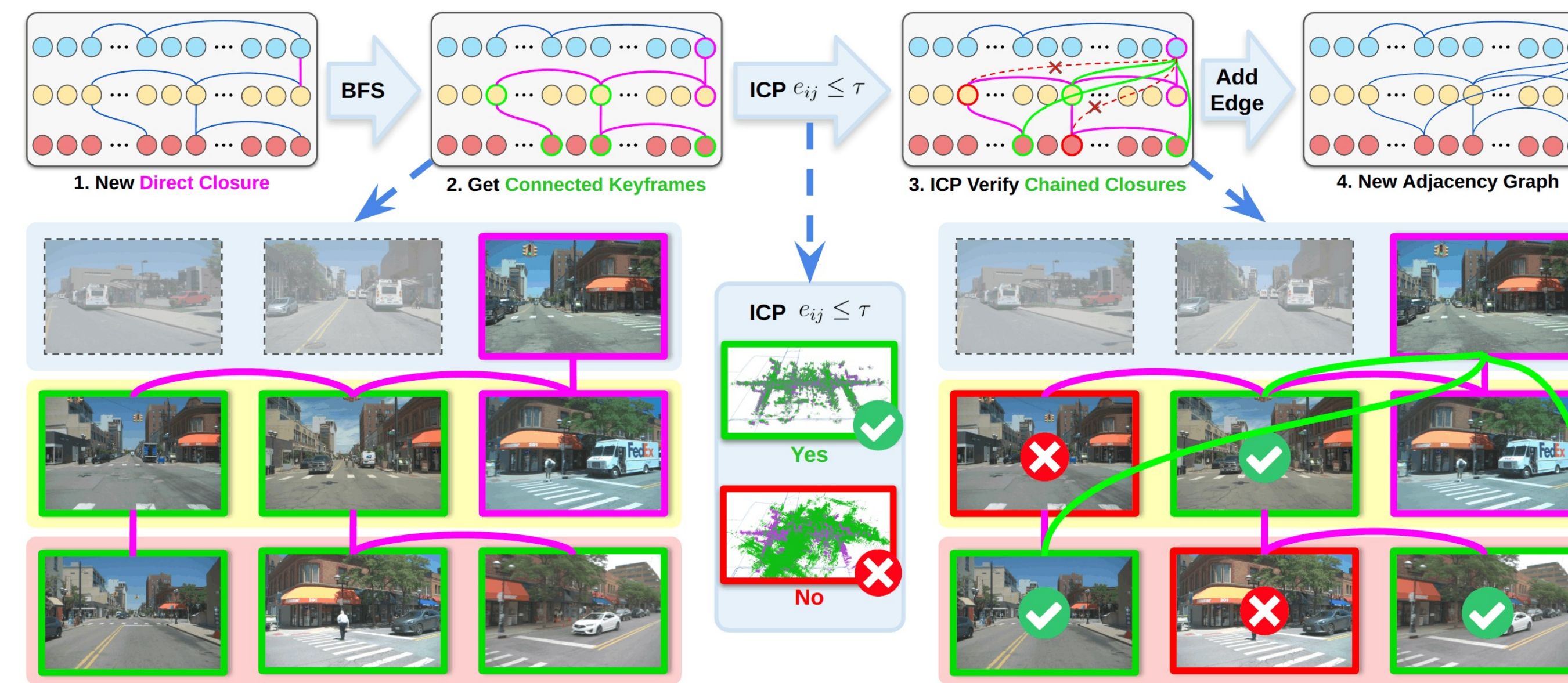
Frontend: a LiDAR-inertial EKF (Fast-LIO2). Backend: one factor graph, all sessions.

Factor graph & loop closure

All sessions share **one graph**, re-solved incrementally by iSAM2. ICP verifies closures; downsampled submaps need no per-LiDAR hyperparam tuning.

Methods

Chained Loop Closure



A direct match to a loaded map frame is a **seed** for many more.

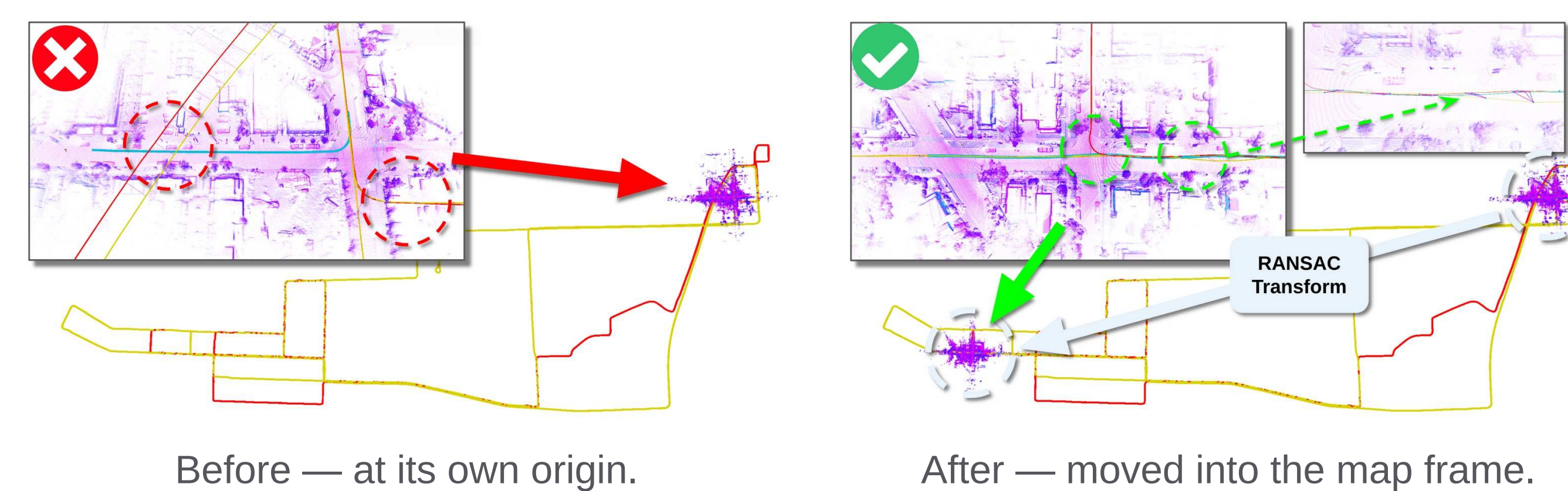
Why it works

- **Efficient**: a graph traversal replaces place recognition.
- **Balanced**: inter-session only; closures never dominate odometry.
- **Robust**: ICP re-verifies each pair, no dynamic object removal needed.

Algorithm

- 1 Direct match j must come from a loaded map: $j < N_L$.
- 2 BFS from j in adjacency graph G gives $N(j)$, loaded frames only.
- 3 For each $k \in N(j)$, ICP keyframe i to submap(k) gives f_{ik} .
- 4 Accept if $f_{ik} \leq \tau$: add factor (i, k) , covariance $\lambda f_{ik} I$.

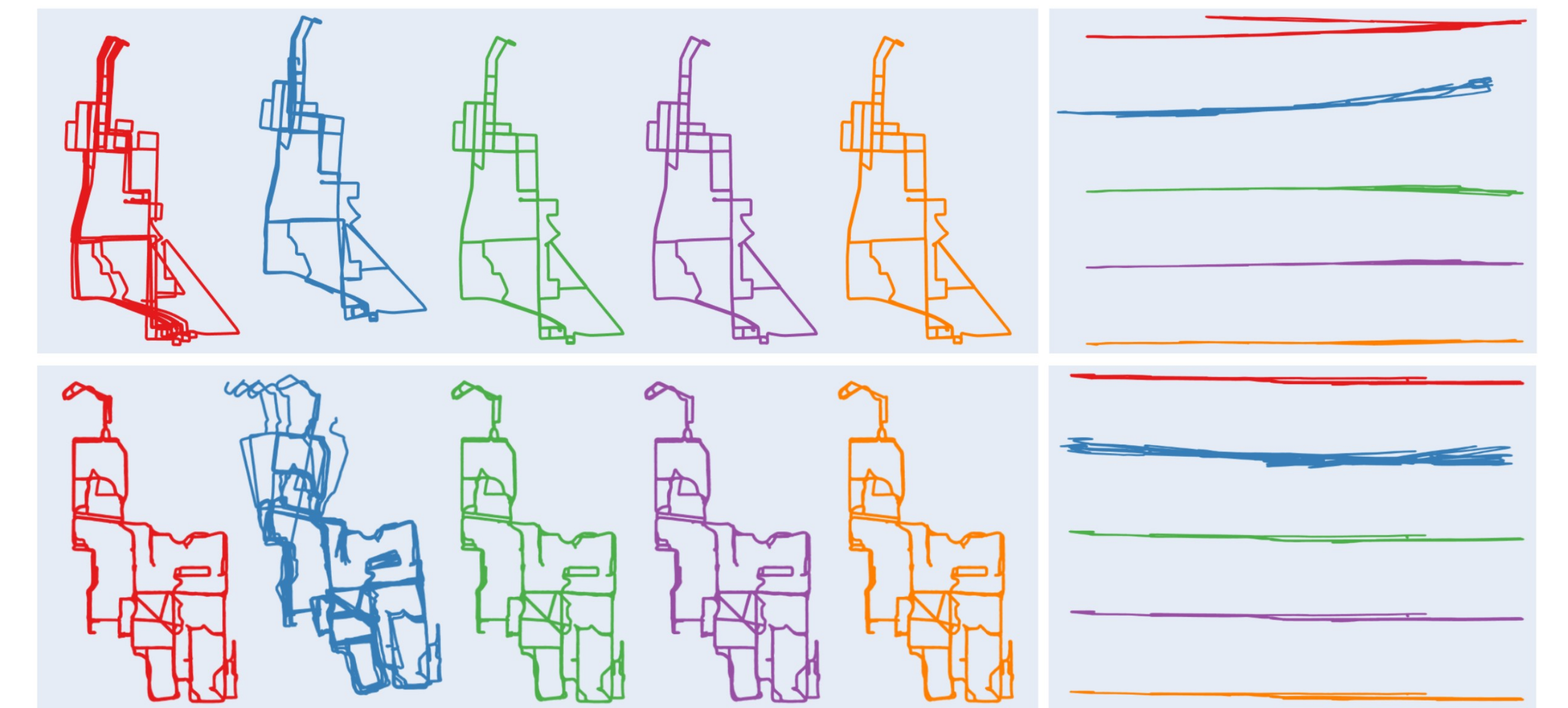
Online Map Merging



- **Propose**: GNSS proximity search in a K-D tree.
- **Register**: FPFH + RANSAC, submap to submap, accepted on fitness.
- **Re-anchor**: swap the prior factor and rebuild the graph.
- **Reuse**: re-detect closures along the aligned trajectory, keep optimizing.

Limitation. GNSS proposes the candidates; LiDAR place recognition would make this GNSS-free.

Results



Top (left) and side (right) view of MARS (upper) and NCLT (lower). Five sessions each.

MARS: 12 h, 1500 × 3000 m, VLS-128, vehicle. NCLT: 8 h, 400 × 800 m, HDL-32E, robot.

Trajectory accuracy: 6-DoF APE RMSE

Method	MARS T (m)	MARS R (°)	NCLT T (m)	NCLT R (°)
Fast-LIO2	52.616	4.708	2.279	2.618
LAMM	43.688	10.285	33.547	11.549
Ours unchained	13.511	3.729	1.473	4.876
Ours chained	9.640	3.448	1.197	4.647

Green = best, yellow = second. Fast-LIO2 is scored per session and averaged.

3D reconstruction: 3DGS test PSNR ↑

Fast-LIO2	LAMM	Ours unchained	Ours chained
11.184	12.444	13.222	13.317

Test rather than train PSNR, since training can overfit on misaligned segments.

Runtime & memory: i9-12900K

Dataset	Method	Frontend (ms)	Backend (ms)	FPS (Hz)	RAM (MB)
MARS	Fast-LIO2	22.28	—	44.88	1065.56
	LAMM	22.28	21.25	22.97	16821.90
	Ours unchained	29.88	402.33	2.31	12571.12
	Ours chained	29.39	447.02	2.10	13668.46
NCLT	Fast-LIO2	7.34	—	136.17	928.77
	LAMM	7.34	11.31	53.61	5917.86
	Ours unchained	7.79	67.50	13.28	5682.82
	Ours chained	7.82	70.13	12.83	5816.96

Runtime over front-end + back-end. Front-end matches Fast-LIO2.

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